

168656

Assignment 2

Part A

Circuit A → series circuit

i) Total resistance

$$20 + 30 + 50 = 100 \Omega$$

ii) Total current

$$I = \frac{V}{R} = \frac{125}{100} = 1.25 A$$

iii) Current through each resistor

since its a series circuit,
current is same

$$= 1.25 A$$

iv) Voltage drop through each resistor

$$V = I \cdot R$$

$$V_1 = 1.25 \times 20 = 25 V$$

$$V_2 = 1.25 \times 30 = 37.5 V$$

$$V_3 = 1.25 \times 50 = 62.5 V$$

v) Power dissipated

$$P = V \cdot I$$

$$P_1 = 25 \times 1.25 = 31.25 W$$

$$P_2 = 37.5 \times 1.25 = 46.875 W$$

$$P_3 = 62.5 \times 1.25 = 78.125 W$$

Circuit B - Parallel

i) Total resistance

$$\frac{1}{R_{\text{total}}} = \frac{1}{20} + \frac{1}{100} + \frac{1}{50} = \frac{8}{100}$$

$$R_{\text{total}} = \frac{100}{8} = 12.5 \Omega$$

ii) Total current

$$I = \frac{V}{R} = \frac{125}{12.5} = 10 \text{ A}$$

iii) Current through each resistor

$$I_1 = \frac{125}{20} = 6.25 \text{ A}$$

$$I_2 = \frac{125}{100} = 1.25 \text{ A}$$

$$I_3 = \frac{125}{50} = 2.5 \text{ A}$$

iv) Voltage drop across each resistor

Since its parallel, voltage is same

$$= 125 \text{ V}$$

v) Power dissipated

$$P = V \cdot I$$

$$P_1 = 125 \times 6.25 = 781.25 \text{ W}$$

$$P_2 = 125 \times 1.25 = 156.25 \text{ W}$$

$$P_3 = 125 \times 2.5 = 312.5 \text{ W}$$